

EXAM 7 – SPRING 2011

2. (4 points)

Given the following loss reserving information as of December 31, 2010:

Accident <u>Year</u>	On-Level Earned <u>Premium</u>	Growth <u>Function</u>	Reported <u>Losses</u>
2008	\$13,500	78.9%	\$7,200
2009	14,000	57.9%	5,700
<u>2010</u>	<u>14,500</u>	13.8%	<u>1,400</u>
Total	42,000		14,300

- Parameter standard deviation for the total estimated unpaid claims is 796.
- The expected accident year loss emergence pattern (growth function) can be approximated by a loglogistic function of the form:

$$G(x | \omega, \theta) = x^\omega / (x^\omega + \theta^\omega),$$

where x denotes time in months from the average accident date to the evaluation date, and G is the growth function describing cumulative percent reported.

- The maximum likelihood estimates of the parameters are:

$$\omega = 1.956 \text{ and } \theta = 15.286.$$

- The actual incremental loss emergence follows an over-dispersed Poisson distribution with scaling factor $\sigma^2 = 9$.

a. (3 points)

Using a truncation point of five years, estimate the total unpaid claims using the Cape Cod method.

b. (1 point)

Calculate the coefficient of variation of the total unpaid claims estimated in part a. above.

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Sample Solutions
Exam 07, 2011, Question 2

Solution 1

a)

	(1)	(2)	(3)	(4)	(5) = (1)×(4)	(6)	ELR = (6)/(5)
AY	On Level Earned Premium	Age at 12/31/2010	Average Age	Growth		Reported Losses	
2008	13,500	36	30	78.9%	10,652	7200	
2009	14,000	24	18	57.9%	8,106	5700	
2010	14,500	12	6	13.8%	2,001	1400	
	42,000				20,759	14,300	68.9%

Growth function for truncation = $x^{\omega} / (x^{\omega} + \Theta^{\omega}) = 54^{1.956} / (54^{1.956} + 15.286^{1.956}) = .9219$

(8) = 0.922 – Est Reserves = (1)
growth × ELR × (8)

13.3%	1,237
34.3%	3,309
78.4%	7,833
	12,378

b) No candidate supplied model answer – from grader model solution.

Process variance = $\text{Sigma}^2 \times E[R] = 9 \times 12,378 = 111,402$

Total Variance = Process Variance + Parameter Variance = $111,402 + 796^2 = 745,018$

Std Dev (Reserves) = $(745,018)^{0.5} = 86$